



L-Theanine – Is It Rest for the Weary?

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If you've spent any time exploring the world of Functional Nutrition, you've probably come across L-theanine. This naturally occurring amino acid has gained attention for its potential role in supporting relaxation, focus, and sleep quality. Unlike many supplements that promise to rev up the body, L-theanine is often described as helping people feel calm, relaxed, and mentally clear without causing drowsiness. [2] *(Not like L-theanine's mirror-image cousin, D-theanine—the Synth lord. Now don't get too attached to that rascal just yet. It may wear the same molecular clothes as L-theanine, but biology can be mighty picky about shape. We'll have ourselves a little chemistry lesson on that in a bit. I promise! Now back to L-theanine...)*

What makes L-theanine especially interesting is that it comes from some surprisingly familiar sources, including green tea, matcha tea, and even a few mushroom species. Researchers continue exploring its effects on the brain and nervous system, [2] while Functional Nutrition practitioners often appreciate its ability to support a balanced stress response. *(I bet I got y'all wondering... what exactly is L-theanine, where can you find it, and what does the science actually say about its benefits? Let's take a closer look.)*

What Is L-Theanine?

(Well Dear, let's read those tender tea leaves and find out!) L-theanine is a naturally occurring amino acid found primarily in tea leaves, particularly green tea, matcha tea, black tea, and a few mushroom species. Unlike many amino acids, it is not used to build proteins in the body. Instead, it appears to act primarily within the nervous system and has become a subject of growing scientific interest. [2]

One of the reasons L-theanine has become increasingly popular is that it appears to promote what researchers describe as a state of "calm alertness"—feeling relaxed without feeling sedated. [2] *(Relaxed but not drugged...I can dig it!)* According to Andrea Nakayama, "L-theanine creates a sense of relaxation and calm about 30 minutes after ingestion. In fact, it stimulates the same alpha waves you activate during meditation. These effects can help counter the jitters that many experience when consuming caffeine." [4] This combination of "calm alertness" has made it a subject of interest for stress management, anxiety support, cognitive function, and sleep quality. [2]

What Is L-Theanine Good For?

- **Stress and Anxiety**

(Modern life seems to keep many of us in a constant state of "go, go, go." Between work, family responsibilities, financial concerns, health challenges, and the never-ending stream of notifications on our phones, it is no surprise that stress-related complaints and health issues



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continue to rise. So, can L-theanine help with my daily stress? Yes, it can!!) Research suggests that L-theanine may help support the body's response to stress by influencing neurotransmitters such as GABA, dopamine, and glutamate. Some studies have demonstrated increased alpha brain-wave activity following L-theanine supplementation, a pattern associated with a relaxed but attentive mental state. [2]

While the overall research remains promising rather than definitive, many individuals report feeling less tense, less “wired,” and better able to cope with everyday stressors after using L-theanine. Importantly, it does not appear to work like a sedative. Rather than shutting the brain down, it may help quiet some of the mental noise. [2,3]

- **Focus and Cognitive Function**

(Focus. Focus. Focus. I need to focus. At first glance, relaxation and focus may seem like opposites. Yet many people discover that when anxiety decreases, concentration improves.)

L-theanine has been studied for its potential effects on attention, reaction time, working memory, and cognitive performance. Researchers believe this may be related to its effects on neurotransmitter balance and its ability to cross the blood-brain barrier. [2]

Interestingly, L-theanine is frequently paired with caffeine. Matcha tea naturally contains both compounds, which may help explain why many people describe matcha as providing a smoother, steadier form of energy compared with coffee. Rather than the “jitter-and-crash” experience that some people report with caffeine alone, the combination may support focused attention with less nervousness. [2,4]

- **Sleep Support**

(Oh, I certainly can use more quality sleep!! L-theanine does this? Wow!!) L-theanine does not appear to function as a traditional sleep aid. Instead, its potential benefit may come from helping calm an overactive mind. Individuals whose sleep difficulties are driven by stress, anxious thoughts, or mental overstimulation may find it easier to relax before bedtime. [1,2]

Several human studies suggest that supplemental L-theanine may improve sleep quality, reduce nighttime awakenings, and promote a greater feeling of refreshment upon waking. However, researchers continue to note that larger and more rigorous studies are still needed. [2]

- **Neurological Health and Urge Incontinence**

Studies show that L-theanine readily crosses the blood-brain barrier and interacts with multiple neurotransmitter systems. Researchers have investigated its effects on glutamate signaling, GABA activity, oxidative stress, neuroinflammation, and neuronal protection. Although much of this work remains preliminary, scientists continue exploring L-theanine's potential role in supporting cognitive function during aging and protecting nervous system health. [2]

One particularly interesting application involves urge incontinence. Unlike stress incontinence—which is often related to pelvic floor weakness—urge incontinence has a significant neurological component. The bladder contracts prematurely, creating a sudden and often



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uncontrollable urge to urinate. Functional Nutrition Alliance educational materials highlight L-theanine as a potential supportive nutrient because of its calming effects on nervous system signaling. [5] Emerging animal research has shown that L-theanine may reduce bladder hyperactivity, improve bladder capacity, and support healthy bladder signaling. While these findings are encouraging, human clinical studies specifically examining L-theanine for urge incontinence remain limited. Therefore, it is best viewed as a promising area of investigation rather than an established treatment. [6,7]

- **Cancer Research (Early and Emerging Evidence)**

Researchers have observed that L-theanine may influence several cellular pathways involved in cancer biology. Some studies suggest it may help enhance the effectiveness of certain cancer therapies while also exhibiting antioxidant and anti-inflammatory properties. However, these findings should be viewed cautiously because much of the evidence comes from cell culture and animal research rather than large human clinical trials. At this time, L-theanine should not be viewed as a treatment for cancer. Rather, it is being investigated as a potentially supportive compound that may one day play a complementary role alongside conventional therapies. [2]

So...What Is GABA?

(Hold your horses... Is it my gabby 2-year-old little sister?) GABA (gamma-aminobutyric acid) is one of the brain's primary calming neurotransmitters. Think of it as part of the body's natural braking system. While excitatory neurotransmitters help keep us alert and active, GABA helps quiet excessive nervous system activity and promotes a sense of calm. So, no we're not talking about your gabby 2-year-old little sister who never stops talking—we ARE talking about a neurotransmitter that helps the brain slow down and relax. [2]

Safety, Risks, and Considerations

L-theanine is generally considered safe and well tolerated when used appropriately. Reported side effects are uncommon but may include headache, dizziness, or gastrointestinal discomfort in some individuals. [1,3]

Because L-theanine may promote relaxation and may have mild blood-pressure-lowering effects, individuals taking medications for blood pressure, anxiety, or sleep should consult their healthcare provider before using supplemental L-theanine. [1,3] Also, research on the use of L-theanine during pregnancy and breastfeeding remains limited. As with many supplements, safety has not been firmly established for these populations. [1] As always, supplements should support a healthy lifestyle and nutrient-dense diet—not replace one.

A Functional Nutrition Perspective

Before reaching for a pill or powder, it is worth considering whether food sources can help meet your health goals. Tea—particularly green tea and matcha—remains the richest and most practical dietary source of L-theanine. While food should always be the first consideration, obtaining therapeutic amounts of a specific nutrient from food alone is not always practical.



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The amount needed to achieve a particular health goal may exceed what most people regularly consume through their diet.

However, there are times when supplementation may be beneficial. For example, green tea naturally contains caffeine and may act as a mild diuretic. Individuals dealing with urinary urgency, urge incontinence, or caffeine sensitivity may find supplementation more practical than increasing tea consumption. [5] As always, Functional Nutrition seeks to identify and address root causes while using foods and supplements strategically to support the body's natural healing processes.

Food First

(Tell me, what foods contain L-theanine?) When it comes to L-theanine, nature provides a few options. Tea remains the richest and most studied dietary source, particularly green tea and *(my favourite)* matcha tea. Smaller amounts have also been identified in certain mushroom species, including the Bay Bolete mushroom. [2] Keep in mind, food sources provide more than a single nutrient. They contain a variety of naturally occurring compounds that may work together synergistically to support health in ways researchers are still working to understand.

What Is Matcha Tea? How Are Those Young Tender Leaves “Babied”?

(Matcha! Matcha tea! I gotta get me some Matcha tea!! -- Sung to the tune of Village People's Macho Man. 😊) Matcha is a special form of green tea made from the youngest and most tender tea leaves. Several weeks before harvest, the tea plants are shaded from direct sunlight. This “babying” process slows growth and encourages the plant to produce higher levels of chlorophyll, amino acids, and other beneficial compounds. The same shading process is believed to also contribute to matcha's naturally higher L-theanine content. [4] *(And honestly Sugar, those little tea leaves are treated better than some folks treat their grandchildren.)* By the way, unlike traditional green tea where the leaves are steeped and discarded, matcha is made by grinding the entire leaf into a fine powder, allowing you to consume all of its nutrients. [4]

Bay Bolete Mushroom: A Forest Scavenger Hunt?

(Y'all we're hunting Bay Boletes today, so bless your heart if you think you're beating me to the pine trees.) The Bay Bolete mushroom (*Imleria badia*) is one of the few foods reported to contain naturally occurring L-theanine. [2] It is found in coniferous and mixed forests throughout parts of North America and Europe and is considered an edible wild mushroom.

The Bay Bolete offers an interesting reminder that God placed beneficial compounds in many unexpected places throughout His creation. For those who enjoy mushroom foraging, this may be one more reason to appreciate a walk through the woods. Just remember—proper identification is essential, as many wild mushrooms have look-alikes that can be harmful *(that's another word for poisonous, y'all!)*.



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Researcher's Note: While Bay Bolete mushrooms contain naturally occurring L-theanine, the concentration can vary considerably based on growing conditions, harvest timing, and preparation methods. Tea remains the most studied, predictable, and practical dietary source of L-theanine.[2]

A Few Words on Supplements

Typical supplemental dosages range from 100–200 mg daily, although some practitioners may recommend higher amounts depending on the individual and the desired outcome. Functional Nutrition Alliance educational materials discuss a trial of 200 mg once or twice daily in the context of neurological support and urge incontinence.[5]

As with any supplement, quality matters. *(So, Sweetie, when you're shopping for theanine, look for "L-theanine" right there on the label. If the label doesn't plainly say 'L-theanine,' I'd keep my wallet right in my pocketbook.)* Supplements should complement—not replace—a nutrient-dense diet.

(Huh? Synthetic mix – The Synth lord? What in tarnation is that?) In the context of L-theanine, the "L" stands for **levorotatory**. Amino acids (except glycine) exist in two spatial forms that are mirror images of one another—a "left-handed" (L-) form and a "right-handed" (D-) form. The "L" indicates the left-handed isomer. This is the naturally occurring form found in tea and the form used in most research studies. [9] In chemistry, some molecules come in mirror-image forms called enantiomers. They contain the same atoms but are arranged differently in space. Since the body is mighty particular about shape, it may respond differently to each form. Theanine found in tea occurs predominantly as the L-form, while synthetic versions may contain a mixture of both L- and D-forms. That's one reason many practitioners recommend choosing products that specifically identify the ingredient as L-theanine and come from reputable manufacturers that verify purity through quality testing. [8,9]

Final Thoughts

Research into L-theanine's effects on brain health and relaxation continues to grow, while emerging evidence suggests potential benefits for neurological health and other areas worthy of further investigation. As encouraging as the research may be, L-theanine is not a magic pill. No supplement can replace the foundations of good health, including nutrient-dense food, quality sleep, movement, stress management, and a healthy relationship with God and others. From a Functional Nutrition perspective, food should always be the first consideration. Fortunately, some of the richest sources of L-theanine come from foods that many people already enjoy, particularly green tea and matcha tea. Supplements may have a place when additional support is needed, but they work best when built upon a strong nutritional foundation.

As always, the goal is not to chase another supplement. The goal is to understand what the body needs, why it needs support, and how we can best provide that support through wisdom, nourishment, and stewardship of the health God has entrusted to us.



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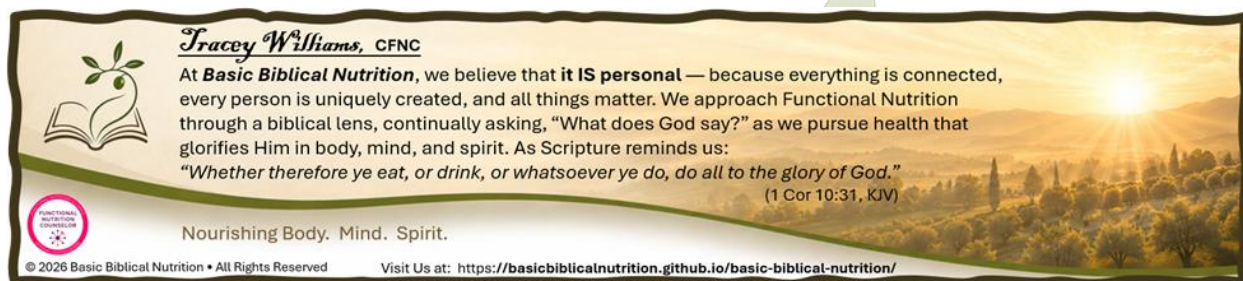
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Tracey Williams is not a medical doctor. She is a Certified Functional Nutrition Counselor/Researcher. Tracey Williams does not diagnose or treat any medical illness or condition.



 **Tracey Williams, CFNC**
At **Basic Biblical Nutrition**, we believe that **it IS personal** — because everything is connected, every person is uniquely created, and all things matter. We approach Functional Nutrition through a biblical lens, continually asking, “What does God say?” as we pursue health that glorifies Him in body, mind, and spirit. As Scripture reminds us:
“Whether therefore ye eat, or drink, or whatsoever ye do, do all to the glory of God.”
(1 Cor 10:31, KJV)

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